

Knowledge acquisition and expert governance

Railway sources
Textbooks and
regulations

Bilingual ingestion
Chunk, align and add
metadata

Governed records
PostgreSQL authoritative
record

Expert review gate
Approve / revise /
reject

Approved knowledge view
Only approved, versioned
records are queryable

status + audit event
approved-only view

revision request

Approved production indexing

Approved text fields
Chinese, English and
provenance

BM25 index
Lexical production index

BGE-M3 encoder
1,024-dimensional vectors

pgvector index
Vectors linked to record IDs

pair-grouped export

retrieval snapshot

Leakage-controlled training and evaluation

Pair-grouped split
ZH/EN pairs stay in one
partition

Train + validation
Approved pairs only

QLoRA training
Completion-only
adapter

Adapter
Local model revision

Evaluation
QA, RAG, translation
and resource use

Held-out test
Evaluation only; never trained or indexed

evaluation only